

Power Supply Systems

Part II

Phase 7: DC Three-Wire System and Unbalanced Loads

Aim: To learn DC three-wire system, neutral current, balancer set and unbalanced-load calculations from absolute zero.

Style: Simple English, SN ma'am style, clean diagrams, no overlapping words, every diagram word explained.

Target: DC 3-wire theory, balanced and unbalanced loading, balancer set, 3-phase 4-wire neutral current and semester-style numericals.

Subject: Power Supply Systems

Section: Part II

Phase: 7 – DC 3-Wire and Unbalanced Loads

Level: Absolute zero to semester-question-ready

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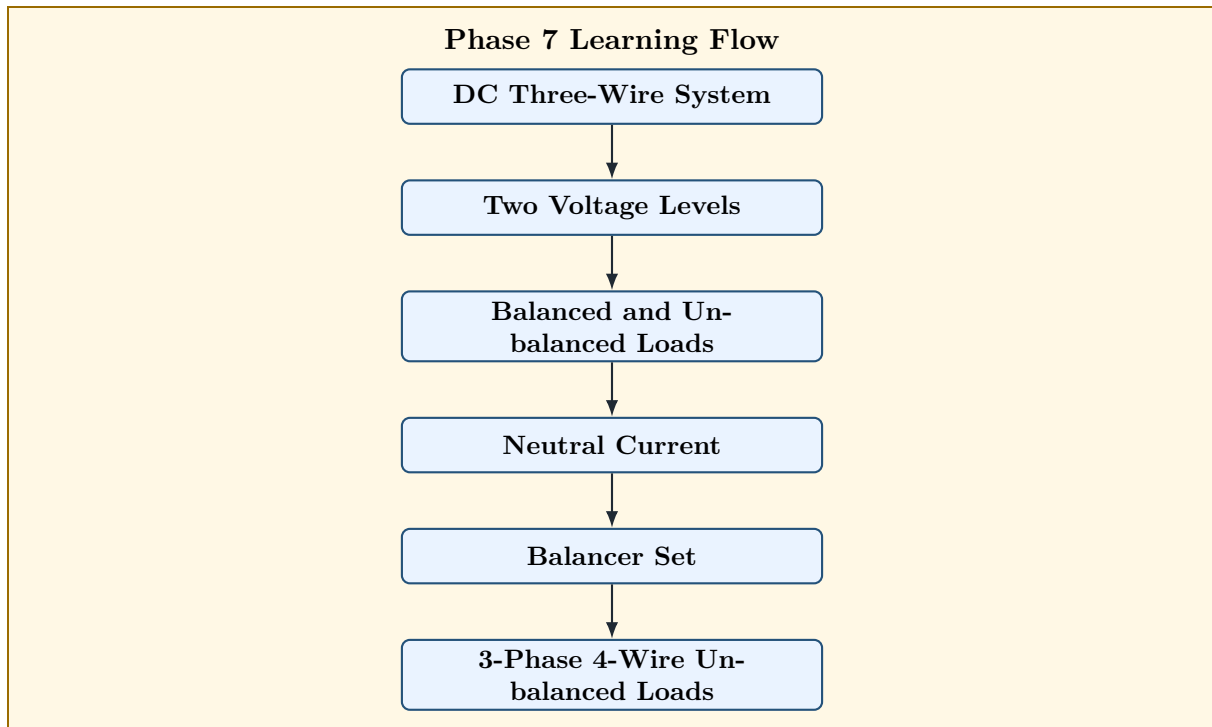
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1 Phase 7 Roadmap

Phase 7 Goal

In Phase 6, we studied ring-main distributor. Now we study another important distribution idea: the **three-wire system**. This phase also prepares us for neutral-current problems in DC and AC systems.



How to Read the Diagram

Word	Meaning
DC Three-Wire System	A DC distribution system using positive outer, neutral and negative outer conductors.
Two Voltage Levels	The system can supply low voltage and double voltage.
Balanced Load	Load on both sides of neutral is equal. Neutral current becomes zero.
Unbalanced Load	Load on the two sides is unequal. Neutral conductor carries the difference current.
Balancer Set	A machine arrangement used to maintain neutral at proper potential.
3-Phase 4-Wire	AC distribution system with R, Y, B phase wires and neutral wire.

2 DC Three-Wire System

Definition

A **DC three-wire system** is a DC distribution system using three conductors: positive outer conductor, neutral conductor and negative outer conductor.

It can supply two different voltages:

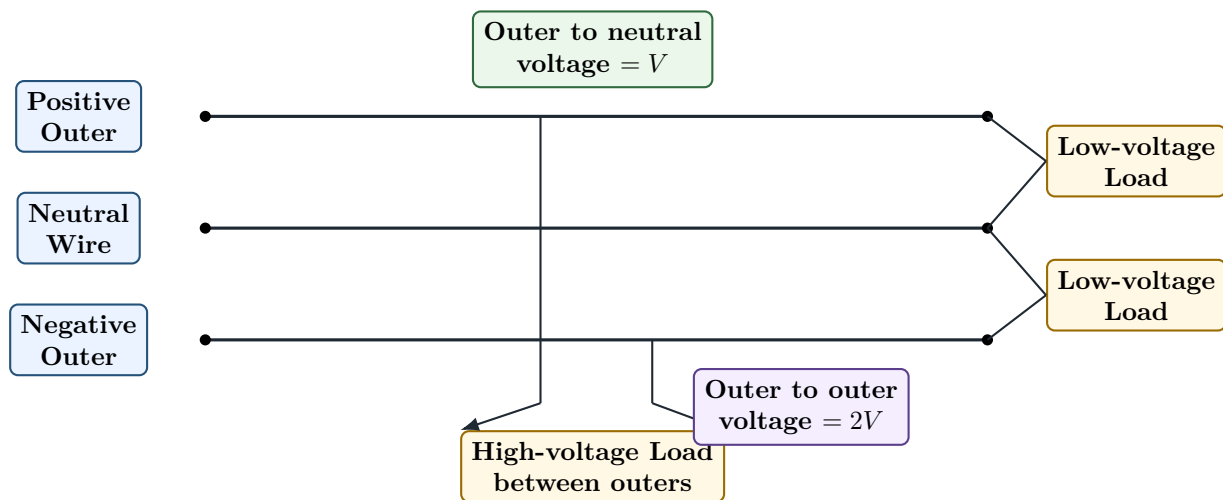
$$\text{Outer to neutral voltage} = V$$

$$\text{Outer to outer voltage} = 2V$$

So, the same system can supply:

- low-voltage loads between one outer and neutral,
- high-voltage loads between the two outer conductors.

3 Basic Diagram of DC Three-Wire System



DC three-wire system supplies both V and $2V$.

How to Read the Diagram

Word/Symbol	Meaning
Positive Outer	The positive conductor of the DC system.
Neutral Wire	The middle conductor. It is at approximately zero or middle potential.
Negative Outer	The negative conductor of the DC system.
Low-voltage Load	Load connected between one outer conductor and neutral. It receives voltage V .
High-voltage Load	Load connected between positive and negative outer conductors. It receives voltage $2V$.
V	Voltage between one outer conductor and neutral.
$2V$	Voltage between the two outer conductors.

4 Advantages of DC Three-Wire System

1. It gives two voltage levels from the same distribution system.
2. It saves conductor material compared with ordinary two-wire system for the same load condition.

3. It is suitable where both lighting and power loads are required.
4. Neutral wire carries no current when the system is balanced.
5. It gives better flexibility of supply.

Semester Answer Style

For a short-note answer, write:

DC 3-wire system gives V and $2V$ using three conductors.

Then draw the diagram and write advantages.

5 Balanced Load in DC Three-Wire System

Definition

A DC three-wire system is said to be **balanced** when the load current on the positive side and the load current on the negative side are equal.

Let:

I_1 = current in positive side load

I_2 = current in negative side load

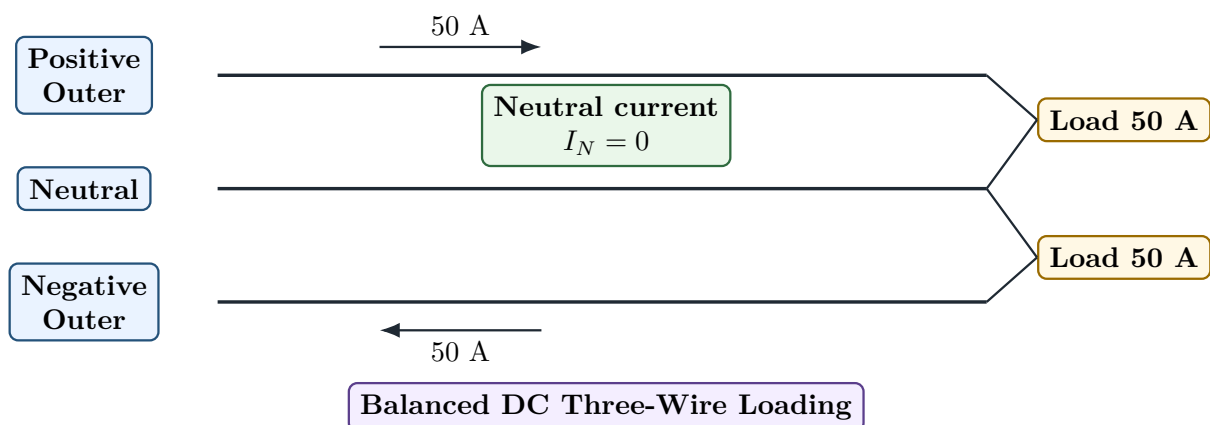
For balanced condition:

$$I_1 = I_2$$

Neutral current:

$$I_N = I_1 - I_2 = 0$$

6 Balanced DC Three-Wire Diagram



How to Read the Diagram

Word/Symbol	Meaning
Load 50 A	Equal current is taken from both sides of the neutral.
Positive Outer	Supplies the upper load.
Negative Outer	Supplies the lower load.
Neutral	Common middle wire. In balanced condition it carries zero current.
$I_N = 0$	Neutral current is zero because upper and lower load currents are equal.
Balanced DC Three-Wire Loading	Load on both sides of neutral is equal.

7 Unbalanced Load in DC Three-Wire System

Definition

A DC three-wire system is said to be **unbalanced** when the load current on one side of the neutral is not equal to the load current on the other side.

If:

$$I_1 \neq I_2$$

then neutral current is:

$$I_N = |I_1 - I_2|$$

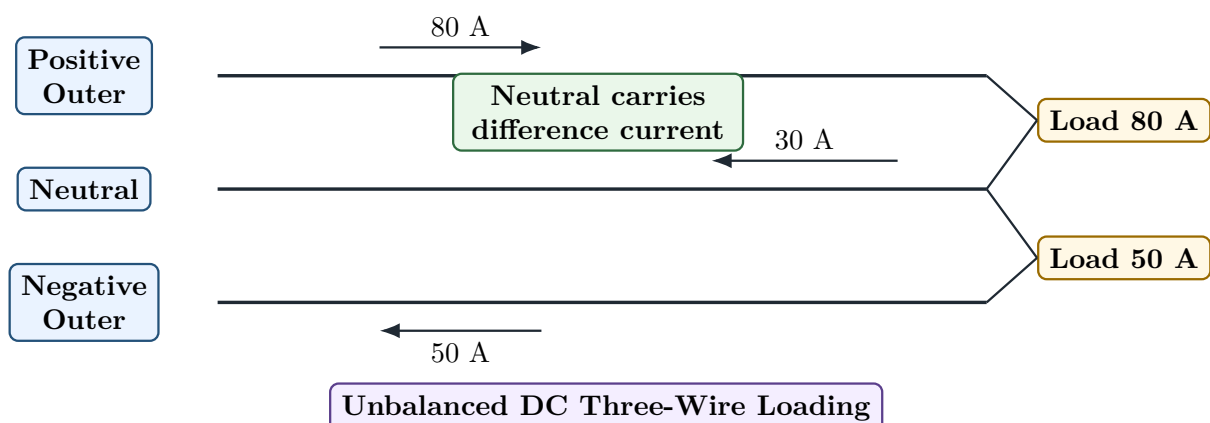
The neutral conductor carries only the difference current.

Memory Hook

In DC three-wire system:

Neutral current = difference of the two side currents

8 Unbalanced DC Three-Wire Diagram



How to Read the Diagram

Word/Symbol	Meaning
Load 80 A	Upper load connected between positive outer and neutral.
Load 50 A	Lower load connected between neutral and negative outer.
Neutral carries difference current	Since $80 \neq 50$, the neutral conductor carries $80 - 50 = 30$ A.
30 A	Neutral current. Its direction depends on which side has larger load current.
Unbalanced DC Three-Wire Load-ing	Load currents on the two sides of neutral are unequal.

9 Solved Example 1: Neutral Current in DC Three-Wire System

9.1 Problem

In a DC three-wire system, the load connected between positive outer and neutral takes 90 A. The load connected between neutral and negative outer takes 65 A. Find the current in the neutral wire.

9.2 Solution

Given:

$$I_1 = 90 \text{ A}$$

$$I_2 = 65 \text{ A}$$

Neutral current:

$$I_N = |I_1 - I_2|$$

$$I_N = |90 - 65|$$

$$I_N = 25 \text{ A}$$

Semester Answer Style

Final sentence: The neutral wire carries only the unbalanced current, which is 25 A.

10 Balancer Set

Definition

A **balancer set** is an arrangement of two identical DC machines connected in series across the two outer conductors, with their common point connected to the neutral wire.

The balancer set is used to keep the neutral point at the correct middle potential.

Balancer set maintains balanced voltage on both sides of neutral.

11 Why Balancer Set is Needed

If the loads on the two sides are unequal, the neutral point may shift. This causes unequal voltages on the two sides:

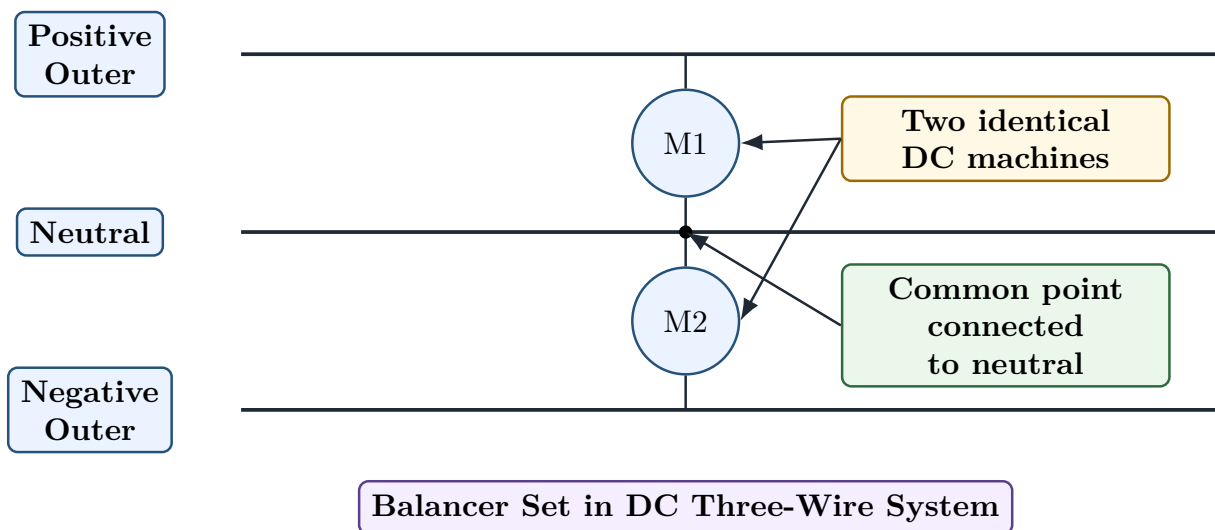
$$V_{+N} \neq V_{N-}$$

This is undesirable because lighting loads or small loads may receive overvoltage or undervoltage.

The balancer set helps by:

- maintaining neutral potential,
- equalising the two side voltages,
- supplying or absorbing the unbalanced current,
- improving voltage balance.

12 Diagram of Balancer Set



How to Read the Diagram

Word/Symbol	Meaning
Positive Outer	Upper conductor of the DC three-wire system.
Neutral	Middle conductor. It should remain at the middle potential.
Negative Outer	Lower conductor of the DC three-wire system.
M1 and M2	Two identical DC machines used as the balancer set.
Two identical DC machines	These machines are connected in series across the two outer conductors.
Common point connected to neutral	The junction of the two machines is connected to the neutral wire.
Balancer Set	The complete arrangement that maintains equal voltage on both sides of neutral.

13 Working of Balancer Set

When the loads on both sides are balanced:

$$I_1 = I_2$$

neutral current is zero and the balancer machines do not need to correct much.

When the loads are unbalanced:

$$I_1 \neq I_2$$

one machine acts like a motor and the other acts like a generator. This action transfers energy from the lightly loaded side to the heavily loaded side.

Memory Hook

Balancer set idea:

One machine acts as motor, the other acts as generator during unbalance.

Its job is to keep neutral voltage balanced.

14 Methods of Obtaining DC Three-Wire System

Common methods are:

1. Two generators connected in series.
2. Three-wire DC generator.
3. Balancer set.
4. Battery system with midpoint connection.

Method	Meaning
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Two generators in series	Two equal DC generators are connected in series and neutral is taken from their junction.
Three-wire DC generator	A special DC generator gives three-wire output.
Balancer set	Two identical DC machines maintain neutral at the middle potential.
Battery midpoint	Neutral point is taken from the midpoint of a battery set.

15 3-Phase 4-Wire System

Definition

A **3-phase 4-wire system** is an AC distribution system having three phase conductors and one neutral conductor.

The four wires are:

$$R, Y, B, N$$

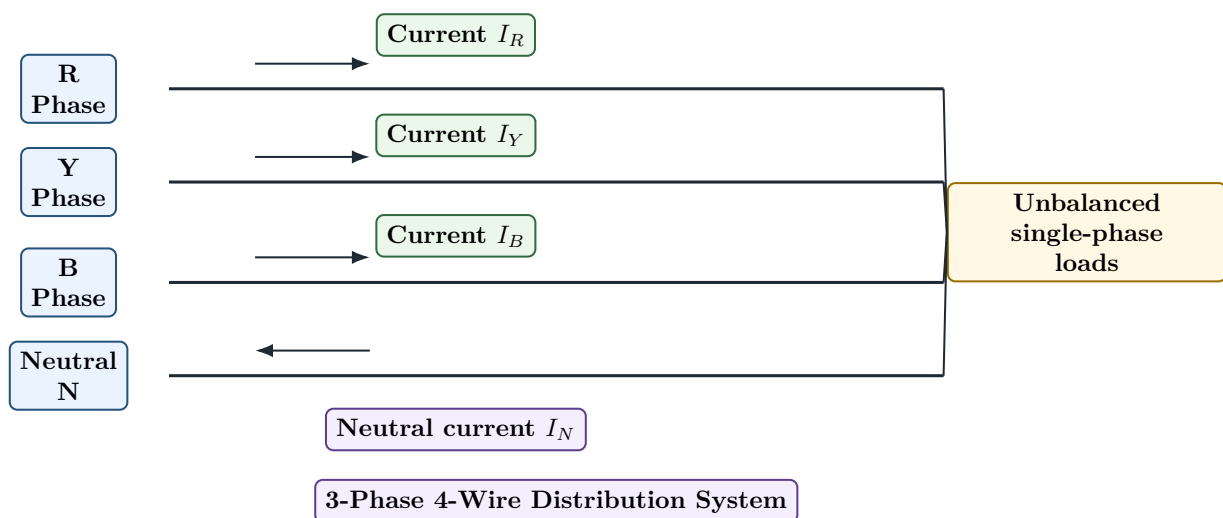
where:

- R, Y, B are three phase wires,
- N is the neutral wire.

It supplies:

- three-phase loads between R, Y, B ,
- single-phase loads between any phase and neutral.

16 Diagram of 3-Phase 4-Wire System



How to Read the Diagram

Word/Symbol	Meaning
R Phase	First phase conductor.
Y Phase	Second phase conductor. It is 120° displaced from R phase.
B Phase	Third phase conductor. It is 120° displaced from the other phases.
Neutral N	Return wire for unbalanced single-phase loads.
Unbalanced single-phase loads	Loads connected from phase to neutral, not necessarily equal in all phases.
I_R, I_Y, I_B	Currents in R, Y and B phase conductors.
I_N	Neutral current, equal to the vector sum of phase currents with opposite sign.

17 Neutral Current in 3-Phase 4-Wire System

For a balanced three-phase load:

$$I_R = I_Y = I_B$$

and the three currents are 120° apart. Their vector sum is zero.

$$I_N = 0$$

For an unbalanced three-phase load:

$$I_R, I_Y, I_B$$

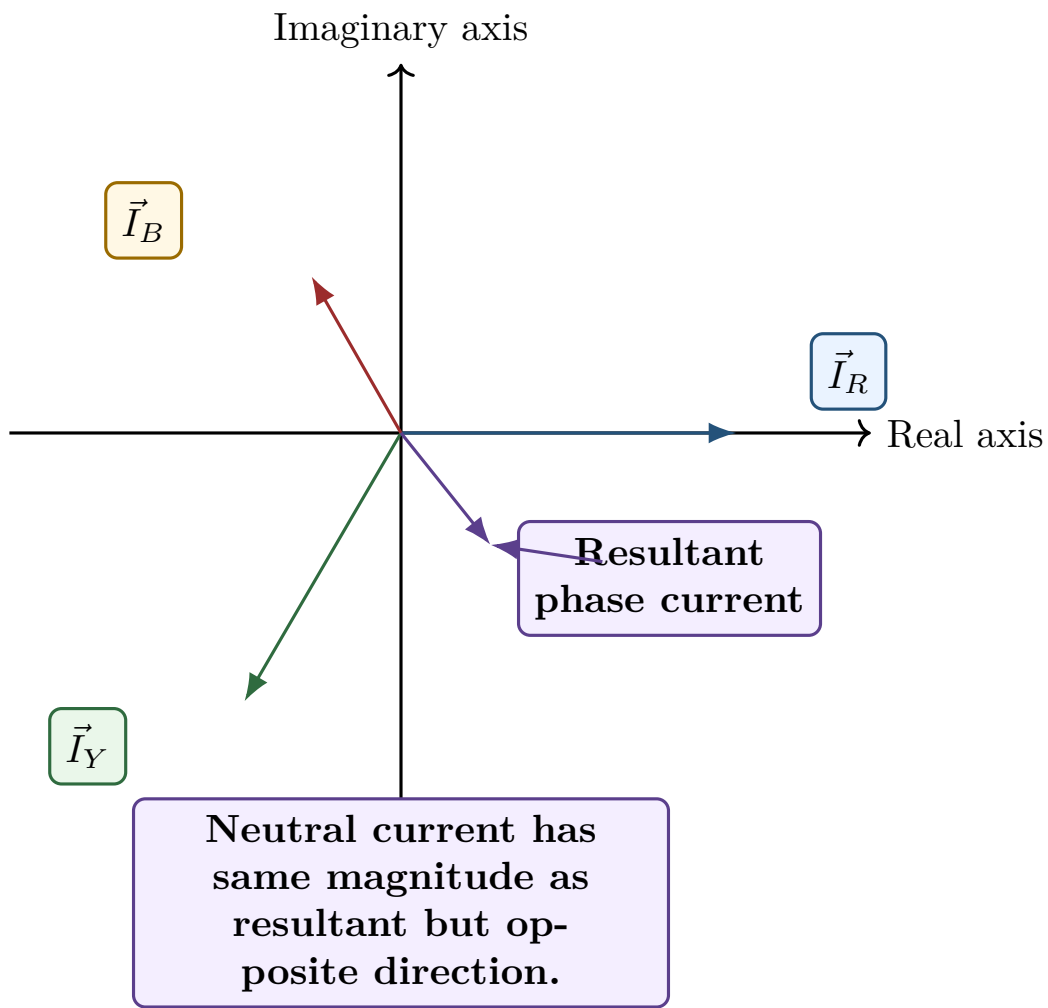
are not equal. So their vector sum is not zero.

$$\vec{I}_N = -(\vec{I}_R + \vec{I}_Y + \vec{I}_B)$$

Magnitude of neutral current:

$$I_N = |\vec{I}_R + \vec{I}_Y + \vec{I}_B|$$

18 Phasor Diagram for Neutral Current



How to Read the Diagram

Word/Symbol	Meaning
Real axis	Horizontal reference axis for phasor calculation.
Imaginary axis	Vertical axis used for rectangular component method.
$\vec{I}_R$	Current phasor of R phase.
$\vec{I}_Y$	Current phasor of Y phase, normally displaced by 120° .
$\vec{I}_B$	Current phasor of B phase, normally displaced by 120° .
Resultant phase current	Vector sum $\vec{I}_R + \vec{I}_Y + \vec{I}_B$.
Neutral current	Equal in magnitude to the resultant but opposite in direction.

19 Formula for Neutral Current

If the three phase currents are 120° apart, then:

$$I_N^2 = I_R^2 + I_Y^2 + I_B^2 + 2I_R I_Y \cos 120^\circ + 2I_Y I_B \cos 120^\circ + 2I_B I_R \cos 120^\circ$$

Since:

$$\cos 120^\circ = -\frac{1}{2}$$

$$I_N = \sqrt{I_R^2 + I_Y^2 + I_B^2 - I_R I_Y - I_Y I_B - I_B I_R}$$

Common Mistake

This formula is valid when the phase currents are separated by 120° . If different power factors are given, use the rectangular component method.

20 Rectangular Component Method

Use this method when phase currents have different power factors or different phase angles.

20.1 Steps

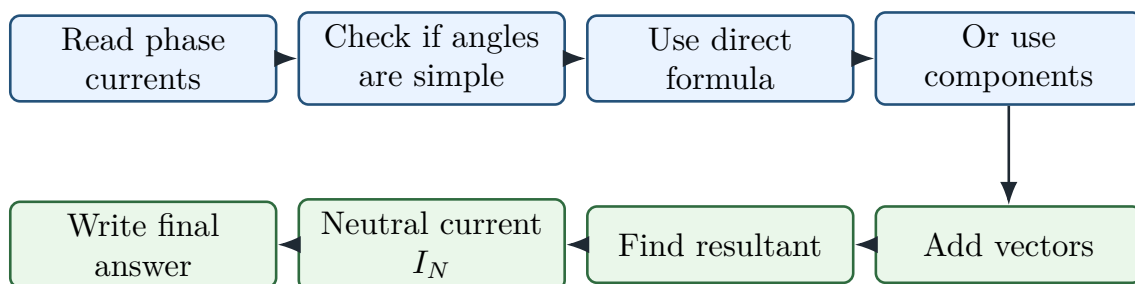
1. Take R-phase voltage as reference.
2. Write each current as a phasor.
3. Resolve each current into horizontal and vertical components.
4. Add all horizontal components.
5. Add all vertical components.
6. Find resultant:

$$I_{\text{resultant}} = \sqrt{(\Sigma I_x)^2 + (\Sigma I_y)^2}$$

7. Neutral current:

$$I_N = I_{\text{resultant}}$$

21 Flowchart for Neutral Current Numerical



How to Read the Diagram

Step	Meaning
Read phase currents	Note I_R , I_Y , I_B from the question.
Check if angles are simple	If currents are balanced by 120° and no different power factor is given, direct formula may be used.
Use direct formula	Use the square-root formula for 120° separated currents.
Or use components	If phase angles are not simple, resolve each current into real and imaginary parts.
Add vectors	Add all phase current vectors.
Find resultant	Resultant magnitude gives neutral current magnitude.
Neutral current I_N	Current flowing in the neutral conductor.
Write final answer	Mention magnitude and, if needed, direction.

22 Solved Example 2: Neutral Current in 3-Phase 4-Wire System**22.1 Problem**

The currents in R, Y and B phases of a 3-phase 4-wire system are:

$$I_R = 30 \text{ A}, \quad I_Y = 20 \text{ A}, \quad I_B = 10 \text{ A}$$

Assuming currents are 120° apart, find the neutral current.

22.2 Solution

Use:

$$I_N = \sqrt{I_R^2 + I_Y^2 + I_B^2 - I_R I_Y - I_Y I_B - I_B I_R}$$

Substitute values:

$$I_N = \sqrt{30^2 + 20^2 + 10^2 - (30)(20) - (20)(10) - (10)(30)}$$

$$I_N = \sqrt{900 + 400 + 100 - 600 - 200 - 300}$$

$$I_N = \sqrt{300}$$

$$I_N = 17.32 \text{ A}$$

Semester Answer Style

Final sentence: The neutral conductor carries 17.32 A due to unbalance in the three phase currents.

23 Solved Example 3: Rectangular Component Method

23.1 Problem

In a 3-phase 4-wire system, currents are:

$$\vec{I}_R = 20\angle 0^\circ \text{ A}$$

$$\vec{I}_Y = 15\angle -120^\circ \text{ A}$$

$$\vec{I}_B = 10\angle 120^\circ \text{ A}$$

Find the neutral current.

23.2 Step 1: Resolve Components

For R phase:

$$I_{Rx} = 20, \quad I_{Ry} = 0$$

For Y phase:

$$I_{Yx} = 15 \cos(-120^\circ) = -7.5$$

$$I_{Yy} = 15 \sin(-120^\circ) = -12.99$$

For B phase:

$$I_{Bx} = 10 \cos(120^\circ) = -5$$

$$I_{By} = 10 \sin(120^\circ) = 8.66$$

23.3 Step 2: Add Components

Horizontal component:

$$\Sigma I_x = 20 - 7.5 - 5 = 7.5$$

Vertical component:

$$\Sigma I_y = 0 - 12.99 + 8.66 = -4.33$$

23.4 Step 3: Resultant Current

$$I_{\text{resultant}} = \sqrt{(7.5)^2 + (-4.33)^2}$$

$$I_{\text{resultant}} = \sqrt{56.25 + 18.75}$$

$$I_{\text{resultant}} = \sqrt{75}$$

$$I_{\text{resultant}} = 8.66 \text{ A}$$

Therefore,

$$I_N = 8.66 \text{ A}$$

Semester Answer Style

Neutral current has the same magnitude as resultant phase-current vector but opposite direction.

24 Comparison: DC Neutral Current and AC Neutral Current

Point	DC Three-Wire System	3-Phase 4-Wire AC System
Conductors	Positive outer, neutral, negative outer	R, Y, B and neutral
Neutral current	Arithmetic difference of two side currents	Vector sum of three phase currents
Balanced condition	$I_1 = I_2$	$I_R = I_Y = I_B$ and 120° apart
Balanced neutral current	0	0
Unbalanced neutral current	$ I_1 - I_2 $	$ \vec{I}_R + \vec{I}_Y + \vec{I}_B $
Main device	Balancer set may be used	Neutral wire carries unbalanced current

25 Reverse Engineering of Semester Questions

Syllabus Word	Concept	Semester Question Form
DC three-wire system	Three conductors and two voltage levels	Explain DC 3-wire system with diagram.
Neutral wire	Carries unbalanced current	Find neutral current for unbalanced loads.
Balanced loading	Equal currents on both sides	Explain why neutral current is zero.
Unbalanced loading	Unequal load currents	Calculate neutral current.
Balancer set	Maintains neutral potential	Explain balancer set with diagram and working.
3-phase 4-wire system	R, Y, B and neutral	Draw diagram and explain neutral current.
Neutral current phasor	Vector sum of phase currents	Find neutral current in unbalanced 3-phase load.

26 Most Predictable Semester Questions from Phase 7

No.	Question	Marks
1	Explain DC three-wire system with a neat labelled diagram.	5
2	What are the advantages of DC three-wire system over two-wire system?	4
3	Explain balanced and unbalanced loading in DC three-wire system.	4–5
4	Find neutral current in a DC three-wire system for given unequal loads.	3–5
5	What is a balancer set? Explain its working with diagram.	5–6
6	Mention methods of obtaining DC three-wire system.	3–4
7	Explain 3-phase 4-wire system with diagram.	4–5
8	Derive or write formula for neutral current in unbalanced 3-phase 4-wire system.	5
9	Calculate neutral current for given R, Y, B phase currents.	5–8
10	Compare neutral current in DC 3-wire and AC 3-phase 4-wire systems.	4–6

27 Exam Answer Templates

27.1 Template 1: DC Three-Wire System

Semester Answer Style

Answer sequence:

1. Define DC three-wire system.
2. Draw three conductors: positive outer, neutral, negative outer.
3. Show low-voltage load between outer and neutral.
4. Show high-voltage load between two outers.
5. Write:

$$V_{\text{outer-neutral}} = V, \quad V_{\text{outer-outer}} = 2V$$

6. Write advantages.

27.2 Template 2: Balancer Set

Semester Answer Style

Answer sequence:

1. Define balancer set.
2. Draw two DC machines connected in series across outers.
3. Connect common point to neutral.
4. Explain that it maintains neutral at middle potential.
5. During unbalance, one machine acts as motor and the other as generator.
6. Write final use: voltage balance and neutral stability.

27.3 Template 3: DC Neutral Current Numerical

Semester Answer Style

Answer sequence:

1. Write two side currents I_1 and I_2 .
2. Use:

$$I_N = |I_1 - I_2|$$

3. Substitute values.
4. Box neutral current.

27.4 Template 4: 3-Phase 4-Wire Neutral Current

Semester Answer Style

Answer sequence:

1. Draw R, Y, B and N system.
2. State:

$$\vec{I}_N = -(\vec{I}_R + \vec{I}_Y + \vec{I}_B)$$

3. If currents are 120° apart, use:

$$I_N = \sqrt{I_R^2 + I_Y^2 + I_B^2 - I_R I_Y - I_Y I_B - I_B I_R}$$

4. If different angles are given, use rectangular components.
5. Box final answer.

28 Phase 7 Final Memory Sheet

One-Page Memory Sheet

1. DC three-wire system has positive outer, neutral and negative outer.

2. It gives two voltage levels: V and $2V$.

3. Balanced DC loading:

$$I_1 = I_2, \quad I_N = 0$$

4. Unbalanced DC loading:

$$I_N = |I_1 - I_2|$$

5. Balancer set maintains neutral at middle potential.

6. Balancer set uses two identical DC machines connected in series.

7. During unbalance, one machine acts as motor and the other as generator.

8. 3-phase 4-wire system has R, Y, B and neutral.

9. Balanced 3-phase neutral current is zero.

10. Unbalanced 3-phase neutral current:

$$\vec{I}_N = -(\vec{I}_R + \vec{I}_Y + \vec{I}_B)$$

11. For 120° separated currents:

$$I_N = \sqrt{I_R^2 + I_Y^2 + I_B^2 - I_R I_Y - I_Y I_B - I_B I_R}$$

29 Phase 7 Self-Test

Answer these without seeing notes:

1. What is a DC three-wire system?
2. What are the three conductors in a DC three-wire system?
3. What two voltage levels are obtained in a DC three-wire system?
4. What is balanced loading?
5. What is unbalanced loading?
6. What is the neutral current in balanced DC loading?
7. What is the neutral current in unbalanced DC loading?
8. What is a balancer set?
9. Why is balancer set used?
10. What are R, Y, B and N in a 3-phase 4-wire system?
11. Why is neutral current zero in balanced 3-phase load?
12. How do you calculate neutral current in unbalanced 3-phase load?

Common Mistake

Do not mix DC neutral-current calculation with AC neutral-current calculation. For DC three-wire:

$$I_N = |I_1 - I_2|$$

For 3-phase 4-wire AC:

$$I_N = \text{vector sum of phase currents}$$

End of Phase 7**Next Phase:****Phase 8 – Fuse and Protection**

Fuse, HRC fuse, prospective current, pre-arcing time, protective relay, circuit breaker and isolator.